C 21545 **(Pages : 4)** **Name.....**

Reg. No.....

FOURTH SEMESTER (CBCSS–UG) DEGREE EXAMINATION, APRIL 2022

Mathematics

MTS 4B 04—LINEAR ALGEBRA

(2019 Admission onwards)

Time : Two Hours and a Half Maximum : 80 Marks

Section A (Short Answer Type Questions)

Answer at least **ten** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 30.

1. Show that the linear system of equations $4x-2y=1$ has infinitely many solutions.
$16x-8y=4$

2. Write any two facts about row echelon forms and reduced row echelon forms.

3. Express the linear system
$$4x_1-3x_3+x_4=1$$
$$5x_1+x_2-8x_4=3$$
$$2x_1-5x_2+9x_3-x_4=0$$
$$3x_2-x_3+7x_4=2$$
in the form AX = B.

4. Let V = R^2 and define addition and scalar multiplication as follows. For $\bar{u}=(u_1,u_2),\bar{v}=(v_1,v_2)$, $\bar{u}+\bar{v}=(u_1+v_1,u_2+v_2)$ and for a real number $k, k\bar{u}=(ku_1,0)$. For $\bar{u}=(1,1)$ and $\bar{v}=(-3,5)$ find $\bar{u}+\bar{v}$ and for $k=5$, find $k\bar{u}$. Also show that one axiom for vector space is not satisfied.

5. Define basis for a vector space.

6. How will you relate the dimension of a finite dimensional vector space to the dimension of its subspace. Give two facts.

7. Give a solution to the change of basis problem.

8. When you can say that a system of linear equation Ax = b is consistent. What is meant by a particular solution of the consistent system Ax = b.

9. Find the rank of a 5 × 7 matrix A for which Ax = 0 has a two-dimensional solution space.

Turn over

10. If $T_A : R^n \to R^m$ is a matrix transformation. Then define its kernel ker (T_A) and Range of (T_A). What is ker (T_A) in terms of null-space of A.

11. Discuss the geometric effect on the unit square of multiplication by a diagonal matrix $A = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}$.

12. Confirm by multiplication that x is an eigen vector of A and find the corresponding eigen value, if $A = \begin{bmatrix} 5 & -1 \\ 1 & 3 \end{bmatrix}$ and $x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

13. Let R^2 have the weighted Euclidean inner product $< u,v > = 2u_1v_1 + 3u_2v_2$. For $u = (1,1), v = (3,2)$, compute $d(u,v)$.

14. If u and v are orthogonal vectors in a real inner product space, then show that $\| u+v \|^2 = \| u \|^2 + \| v \|^2$.

15. State four properties of orthogonal matrices.

(10 × 3 = 30 marks)

Section B (Paragraph/ Problem Type Questions)

Answer at least **five** *questions.*
Each question carries 6 *marks.*
All questions can be attended.
Overall Ceiling 30.

16. Suppose that the augmented matrix for a linear system has been reduced to the row echelon form as $\begin{bmatrix} 1 & 0 & 8 & -5 & 6 \\ 0 & 1 & 4 & -9 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{bmatrix}$ solve the system.

17. If A is an invertible matrix, then show that A^T is also invertible and $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$.

18. Let V be a vector space and $\overline{u}$, a vector in V and k a scalar. Then show that (i) $O\overline{u} = 0$; (ii) $(-1)\overline{u} = -\overline{u}$.

19. If $S=[v_1,v_2,\ldots,v_n]$ is a basis for a vector space V, then show that every vector v in V can be expressed in form $v=c_1v_1+c_2v_2+\ldots+c_nv_n$ in exactly one way. What are the co-ordinates of V relative to the basis S ?

20. Consider the basis $B=[u_1,u_2]$ and $B'=[u_1',u_2']$ for R^2, where $u_1=(2,2)$ $u_2=(4,-1)$ $u_1'=(1,3)$ $u_2'=(-1,-1)$.

    (a) Find the transition matrix from B' to B.

    (b) Find the transition matrix from B to B'.

21. If A is a matrix with n columns, then define rank A, nullity of A and establish a relationship between them.

22. Define eigen space corresponding to an eigen value λ of a square matrix A. Also find eigen value and bases for the eigen space of the matrix $A=\begin{bmatrix}1 & -2\\ 0 & 1\end{bmatrix}$.

23. Use the Gram–Schmidt process for an orthonormal basis corresponding to the basis vectors $u_1=(1,1,1)$, $u_2=(0,1,1)$ and $u_3=(0,0,1)$.

(5 × 6 = 30 marks)

Section C (Essay Type Questions)

Answer any **two** *questions.*
Each question carries 10 *marks.*

24. Show that the following statements are equivalent for an $n \times n$ matrix A :

    (a) A is invertible.

    (b) $Ax = 0$ has only the trivial solution.

    (c) The reduced row echelon form of A is I_n.

    (d) A is expressible as a product of elementary matrices.

25. (a) Define Wronskian of the functions $f_1=f_1(x),f_2=f_2(x)\ldots f_n=f_n(x)$ which are $n-1$ times differentiable in $(-\infty,\infty)$. Use this to show that $f_1=x$ and $f_2=\sin x$ are linearly independent vectors in $c^{\infty}(-\infty,\infty)$.

    (b) Show that the vectors $v_1=(1,2,1),v_2=(2,9,0)$ and $v_3(3,3,4)$ form a basis for R^3.

Turn over

26. (a) If A is the matrix $\begin{bmatrix} 1 & -2 & 0 & 0 & 3 \\ 2 & -5 & -3 & -2 & 6 \\ 0 & 5 & 15 & 10 & 0 \\ 2 & 6 & 18 & 8 & 6 \end{bmatrix}$, then find a basis for the row space consisting on entirely row vectors from A.

(b) Find the standard matrix for the operator $T : R^3 \to R^3$ that first rotates a vector counter clockwise about z-axis through an angle θ, reflects the resulting vector about yz plane and then projects that vector orthogonally onto the xy plane.

27. (a) On P_2, polynomial in $[-1,1]$, define innerproduct as $< p,q > = \int_{-1}^{1} p(x)q(x)dx$. Find $\| p \|, \| q \|$ and $< p,q >$ for $p = x$ and $q = x^2$.

(b) If A is an $n \times n$ matrix with real entries, show that A is orthogonally diagonalizable if and only if A has an orthonormal set of n eigenvectors.

(2 × 10 = 20 marks)

C 21545–A (Pages : 4) Name.....

Reg. No.....

FOURTH SEMESTER (CBCSS–UG) DEGREE EXAMINATION, APRIL 2022

Mathematics

MTS 4B 04—LINEAR ALGEBRA

(2019 Admission onwards)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes	Total No. of Questions : 20	Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 4B 04—LINEAR ALGEBRA

(Multiple Choice Questions for SDE Candidates)

1. If A and B are square matrices of the same order, then :
   $(AB)^T$ =
   (A) $A^T B^T$. (B) $B^T A^T$.
   (C) $A^T + B^T$. (D) $(BA)^T$.

2. A matrix that is both symmetric and upper triangular must be a :
   (A) Diagonal matrix. (B) Non-diagonal but symmetric.
   (C) Both (A) and (B). (D) None of the above.

3. If A and B are invertible matrices with the same size, then AB is invertible and $(AB)^{-1}$ =.
   (A) $A^{-1}B^{-1}$. (B) $B^{-1}A^{-1}$.
   (C) Both A and B. (D) None of the above.

4. A matrix E is called ———— if it can be obtained from an identity matrix by performing a single elementary row operation.
   (A) Equivalent matrix. (B) Echelon matrix.
   (C) Elementary matrix. (D) Row reduced matrix.

5. A homogeneous linear system in n unknowns whose corresponding augmented matrix has a reduced row echelon form with r leading 1's has
   (A) n-free variables (B) $n - r$ free variables.
   (C) r-free variables. (D) Cannot be determined.

6. A consistent linear system of two equations in two unknowns has :
   (A) Exactly one solution. (B) Infinitely many solutions.
   (C) Exactly two solutions. (D) Either (A) or (B).

7. If $T_A : R^n \to R^m$ and $T_B : R^n \to R^m$ are matrix transformations, and if $T_A(x) = T_B(x)$ for every vector x in R^n, then :
   (A) (A) and (B) are equivalent but not equal.
   (B) (A) and (B) are equal.
   (C) (A) and (B) cannot be equal.
   (D) Cannot be determined.

8. If W is a subspace of a finite-dimensional vector space V, then :

(A) dim (W) = dim (V) always. (B) dim (W) ≥ dim (V).

(C) dim (W) ≤ dim (V). (D) None of the above.

9. A $n \times n$ matrix has :

(A) At most n distinct eigenvalues. (B) At least n distinct eigenvalues.

(C) Exactly n distinct eigenvalues. (D) Exactly $n + 1$ distinct eigenvalues.

10. Find the value of m such that the vector $(m, 7, -4)$ is a linear combination of vectors $(-2, 2, 1)$ and $(2, 1, -2)$:

(A) 2. (B) – 2.

(C) 0. (D) – 1.

11. Suppose that $x = (2, 1, 0, 3)$, $y = (3, -1, 5, 2)$. and $z = (-1, 0, 2, 1)$. Which of the following vectors are in *span* $\{x, y, z\}$?

(A) (2, 3, – 7, 3). (B) (1, 1, 1, 1).

(C) Both (A) and (B). (D) Neither (A) nor (B).

12. Which of the following is true ?

(A) A finite set that contains 0 is linearly dependent.

(B) A set with exactly one vector is linearly independent if and only if that vector is not 0.

(C) A set with exactly two vectors is linearly independent if and only if neither vector is a scalar multiple of the other.

(D) All are true.

13. Which of the following sets of vectors in R3 are linearly independent.

(A) {(2, 1, 2), (8, 4, 8)}.

(B) {(1, 1, 0), (1, 1, 1), (0, 1, –1)}.

(C) {(1, 3, 2), (1, – 7, – 8), (2, 1, –1)}.

(D) {(– 2, 0, 1), (3, 2, 5), (6, – 1, 1), (7, 0, –2)}.

Turn over

14. Transition matrices are :

(A) Not at all invertible. (B) Invertible always.

(C) Invertible sometimes. (D) Data not complete.

15. Let A be any matrix. Then :

(A) *rank* (A) = *rank* (A^T). (B) *rank* (A) $\neq$ *rank* (A^T).

(C) *rank* (A) < *rank* (A^T). (D) *rank* (A) > *rank* (A^T)..

16. If A is an $m \times n$ matrix, then :

(A) The null space of A and the row space of A are orthogonal complements in R^n.

(B) The null space of A^T and the column space of A are orthogonal complements in R^m.

(C) Both (A) and (B) are correct.

(D) Neither (A) nor (B) are correct.

17. Let A is an $n \times n$ matrix. The eigenspace of A corresponding to λ is same as :

(A) The null space of the matrix $\lambda I - A$.

(B) The kernel of the matrix operator $T_{\lambda I - A} : R^n \to R^n$

(C) The set of vectors for which $Ax = \lambda x$.

(D) All the above.

18. Let A is an $n \times n$ matrix and suppose A has rank n. Then :

(A) T_A is not one-to-one. (B) $\lambda = 0$ is not an eigenvalue of A.

(C) The range of T_A is {0}. (D) The kernel of T_A is R^n.

19. Which of the following is true ?

(A) $(u, v + w) = \langle u, v \rangle + \langle u, w \rangle$. (B) $\langle u, v + w \rangle = \langle v, u \rangle + \langle w, u \rangle$.

(C) Both (A) and (B) are true. (D) Neither (A) nor (B) is true.

20. Find the correct one from the given statements :

(A) If u is orthogonal to every vector of a subspace W, then $u = 0$.

(B) If u and v are orthogonal, then $|\langle u, v \rangle| = \|u\| \, \|v\|$.

(C) If u and v are orthogonal then $\| u + v \| = \|u\| \, \|v\|$.

(D) None of these.